ASSIGNMENT -1 -CO1

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COURSE:	**MOBILE COMPUTING**

TASK 1 – Design Alternative Frequency-Planning Strategies

A fixed spectrum can be reused in different geographical areas. The same channels are assigned to cells that are sufficiently far apart so that their signals do not create unacceptable co-channel interference. Three distinct strategies are considered.

Strategy 1: Fixed Cluster Frequency Reuse (FCA)

The service area is divided into repeating clusters of N hexagonal cells. Each cell in a cluster receives a different channel group. The complete channel pattern repeats in the next cluster. For N=16, the 336 channels are divided into 16 groups of 21 channels each. This gives predictable planning and simple operation.

- Access-channel concept: in FDMA, each call receives a separate frequency pair; in TDMA, users share a carrier in different time slots; in OFDMA, users receive groups of subcarriers and resource blocks. Control channels carry access, paging and signalling; traffic channels carry voice/data.
- Assumptions: approximately regular hexagonal cells, planned antenna coverage, stable channel groups and a sufficiently uniform initial traffic map.
- Design principle: spatial reuse. The same channel group is reused only after the required reuse distance.
- Why investigate: it is simple, predictable and provides a baseline against which more flexible strategies can be compared.

Iterative pseudocode:

```
INPUT: total_channels=336, candidate cluster sizes N, hexagonal grid\nFOR each valid N = i*i + i*j + j*j:\n compute Q = sqrt(3*N) and approximate SIR\n    IF SIR >= 12 dB:\n        divide 336 channels into N groups\n assign one group to each cell position in the cluster\n        repeat the cluster pattern over the grid\n calculate channels per cell and Erlang-B capacity\nOUTPUT: feasible N values and selected channel map
```

Strategy 2: Sectorised Frequency Reuse

Each base station is divided into directional sectors, commonly three 120-degree sectors or six 60-degree sectors. Directional antennas concentrate energy towards the intended sector and reduce the number of strong co-channel interferers seen by a user. The same site can therefore reuse spectrum more efficiently, subject to antenna isolation and side-lobe leakage.

- Access channels are assigned to each sector. Control and traffic resources are planned per sector; the air-interface method may be FDMA/TDMA or OFDMA depending on the radio system.
- Assumptions: directional antenna patterns are installed correctly, sector boundaries are surveyed and neighbouring sectors are coordinated.
- Design principle: spatial filtering. Antenna direction reduces interference from unwanted directions.
- Why investigate: it can improve SIR without acquiring additional spectrum, especially in busy academic and stadium zones.

Pseudocode:

```
FOR each base-station site:\n    divide the coverage area into 3 or 6 sectors\n    assign a coordinated channel/resource group to each sector\n    FOR each user location:\n        identify serving sector and strongest interfering sectors\n        estimate received wanted and interfering power using antenna gains\n        calculate SIR and blocking\n    adjust sector azimuth, tilt and power until coverage targets are met
```

Strategy 3: Dynamic Channel Assignment and Borrowing (DCA)

Instead of permanently fixing every channel to one cell, a central controller or distributed radio controller assigns an available channel when a call arrives. A lightly loaded neighbouring cell may lend a channel to a busy cell, provided the borrowing does not violate interference constraints. When the call ends, the channel returns to the pool.

- Access channels are selected from a common or partially common pool. The controller checks availability, reuse distance, interference and handoff reservations before assigning a traffic channel.
- Assumptions: cells exchange load information, channel occupancy is updated quickly and a borrowing decision can be reversed safely.
- Design principle: trunking efficiency and adaptive resource allocation.
- Why investigate: traffic is not uniform. Hostels, lecture halls and stadiums can be busy at different times, so dynamic allocation can reduce blocking compared with rigid channel partitions.

Pseudocode:

```
WHEN a call request arrives in cell c:\n    list channels free in c and eligible neighbouring cells\n    reject channels that violate reuse/interference constraints\n    choose the eligible channel with the lowest predicted interference\n    if none is available, queue or block the request\n    otherwise assign channel and update occupancy\nWHEN call ends: release channel and update the pool
```

Preprocessing and cost responsibility

Before any strategy is selected, the operator needs a land-use and traffic map, building-height information, a propagation survey, candidate site locations, a hexagonal grid, road/access information and an estimate of busy-hour demand by zone. The operator normally bears the cost of radio planning, drive/walk testing, traffic surveys, engineering design and site acquisition. The university or township authority may assist with access to rooftops, poles and land, but the licensed operator remains responsible for network performance.

TASK 2 – Identify the Governing Design Parameter

Strategy	**Main governing parameter**	**Why it governs capacity/interference**	**Growth response**
Fixed cluster reuse	Cluster size N (and $Q=\sqrt{3N}$)	N controls channels per cell S/N and reuse distance; larger N improves SIR but reduces channels per cell.	Choose the smallest N that satisfies SIR; add cells or reduce R as demand rises.
Sectorisation	Number of sectors and effective interferer count	More directional sectors reduce strong interferers but increase sector boundaries and handoff complexity.	Increase sectorisation in concentrated high-load zones where antenna isolation is practical.
Dynamic assignment	Available channel pool and per-cell offered load A	The controller can use idle channels from other cells while preserving interference constraints.	More frequent load-aware reassignment and better coordination are needed at high demand.

For fixed reuse, N is the key trade-off. Increasing N increases reuse distance and SIR, but each cell receives fewer channels. For sectorisation, the effective number and strength of interferers are important. For dynamic assignment, the ratio of offered traffic to available channels in each cell is important.

TASK 3 – Derive Capacity and Interference Functions Mathematically

3.1 Hexagonal geometry and cluster size

A regular hexagonal cell has six neighbouring cells. Moving between co-channel cells on a hexagonal grid by i cells in one direction and j cells at 60 degrees gives the cluster-size relation $N=i^2+ij+j^2$. Examples include N=3 (i=1,j=1), N=4 (2,0), N=7 (2,1), N=12 (2,2), and N=16 (4,0).

The centre-to-centre reuse distance is $D=R\sqrt{(3N)}$. Therefore $Q=D/R=\sqrt{(3N)}$. For N=16, $Q=\sqrt{48}=6.928$ and $D=6.928R$.

The area of a regular hexagon with centre-to-vertex radius R is $A_{cell}=(3\sqrt{3}/2)R^2=2.598R^2$. The number of cells required to cover area A is approximately C=ceil(A/A_cell), before adding overlap, edge coverage and reserve sites.

For R=0.5 km, $A_{cell}=2.598\times0.5^2=0.6495$ km². Thus the geometric minimum for 12 km² is ceil(12/0.6495)=19 cells. This is only a coverage estimate; capacity requires more cells.

3.2 Channels per cell and Erlang-B traffic capacity

For N=16, channels per cell m=S/N=336/16=21. Erlang-B is used because calls that arrive when all channels are occupied are blocked. Repeatedly applying $B(m,A)=A\cdot B(m-1,A)/(m+A\cdot B(m-1,A))$ gives a maximum offered load of approximately 14.036 Erlangs for 21 channels at 2% blocking. This value is an approximate planning limit per cell.

Subscribers	**Total offered traffic**	**Approx. cells needed at 14.036 E/cell**	**Comment**
1,000	25 Erlangs	2	Traffic is low; coverage and minimum site spacing dominate.
10,000	250 Erlangs	18	Capacity begins to require more cells than the initial coverage grid.
20,000	500 Erlangs	36	Additional capacity sites are required.
100,000	2,500 Erlangs	179	Dense small-cell deployment or other capacity enhancement is needed.

Calculation example for 100,000 subscribers: $A_{total}=100{,}000\times0.025=2{,}500$ Erlangs. Required cells=ceil(2,500/14.036)=179. The approximate cell radius needed to fit 179 equal hexagons into 12 km² is

R≈sqrt(12/(179×2.598))=0.161 km. This is a first-order estimate and does not include overlap, edge effects, uneven demand or reserve capacity.

3.3 Co-channel SIR derivation

Assume the wanted mobile is at distance R from its serving base station and the six first-tier co-channel base stations are each approximately D away. With path loss proportional to distance raised to n, wanted received power is proportional to R^(−n), while each interfering power is proportional to D^(−n). The total interference is approximately 6D^(−n). Hence SIR≈R^(−n)/[6D^(−n)]=(D/R)^n/6=(3N)^(n/2)/6.

For N=16: Q=√48=6.928. At n=2.5, SIR=48^1.25/6=21.06, or 13.23 dB. At n=3, SIR=48^1.5/6=55.43, or 17.44 dB. At n=4, SIR=48²/6=384, or 25.84 dB. The conservative open-campus estimate is therefore above the selected 12 dB minimum under the six-equal-interferer approximation.

Sectorisation changes the SIR denominator. If directional filtering reduces the number and power of dominant interferers, then SIR≈wanted power divided by the sum of the remaining weighted interferers. It is not correct to assume that sectorisation always divides the interferer count by three exactly; actual antenna patterns, side lobes and geometry must be measured.

Dynamic channel assignment does not automatically change the physical SIR formula. It selects only channels that pass the reuse/interference check. Its benefit is improved use of idle resources while maintaining the required SIR.

3.4 Planning cost and cost–capacity relation

Let C_site be the one-time cost of a site, C_survey the planning and measurement cost, C_sector the extra sector equipment cost, and C_control the controller/software cost. If M sites are installed, a simple planning-cost model is C_total=C_survey+M·C_site+M·C_sector+C_control. For fixed reuse, capacity is approximately M·A_B, where A_B is the Erlang-B traffic supported per cell. Thus cost per supported Erlang is C_total/(M·A_B).

FCA has lower control complexity but may need more sites when traffic is uneven. Sectorisation adds antenna and radio cost but may improve SIR and capacity per site. DCA adds controller and coordination cost but can reduce unused channel capacity in lightly loaded cells. Actual currency estimates cannot be calculated because the assignment does not provide equipment, rent, power or survey prices.

TASK 4 – Analyse Best, Average and Worst-Case Behaviour

SIR depends on mobile position. Near the serving base station, the wanted signal is strong and SIR is normally best. At mid-cell, wanted power is lower and SIR is moderate. Near a cell edge or corner, the wanted signal is weaker and the mobile may be close to several neighbouring cells, so SIR is normally worst.

Case	Mobile position / traffic	SIR and blocking explanation

Best	Near serving base station; off-peak	Short wanted-signal distance gives high received power. Few channels are occupied, so blocking is close to zero.
Average	Typical locations; average busy hour	Wanted and interfering powers are moderate. Erlang-B blocking approaches the design target only when the cell is loaded near its planned limit.
Worst	Cell edge/corner; event surge	Wanted signal is weak and multiple neighbours may interfere. High offered load fills channels and blocking rises sharply.

Example: for N=16 and n=2.5, the ideal six-interferer SIR estimate is 13.23 dB. This is a cell-edge planning approximation. Near the base station the wanted distance is smaller than R, so SIR increases. If shadowing reduces the wanted signal by 6 dB, the effective SIR may also fall by roughly 6 dB; therefore a design with only 1.23 dB margin above the 12 dB threshold needs field validation.

The busy-hour offered traffic for 1,000 subscribers is 25 E, for 10,000 it is 250 E, for 20,000 it is 500 E and for 100,000 it is 2,500 E. Blocking is not determined by subscriber count alone; it depends on how traffic is distributed among cells and the number of channels in each cell. A local event can overload one zone even when the township-wide average is acceptable.

A larger path-loss exponent reduces distant co-channel interference more rapidly, improving the ideal SIR estimate. However, dense buildings can also create shadowing, penetration loss and coverage holes. Therefore n=4 does not automatically mean better user experience; it can mean weaker coverage behind obstacles.

TASK 5 – Scaling and Sensitivity Analysis

For fixed reuse, channels per cell are S/N, so the number of channels available per cell is proportional to 1/N. Under the six-interferer approximation, SIR is proportional to N^(n/2). For fixed township area, the number of cells is proportional to $1/R^2$. Offered traffic is proportional to the number of subscribers.

Change	**Mathematical effect**	**Planning interpretation**
N increases	Channels/cell $\propto$ 1/N; SIR $\propto$ N^(n/2)	Better interference protection but fewer channels in each cell.

R decreases	Cell count $\propto 1/R^2$	More sites and higher aggregate capacity, but more handoffs and site cost.
Subscribers increase	Traffic $\propto$ subscribers	Capacity must rise through smaller cells, sectorisation and/or dynamic assignment.
n rises from 2.5 to 4	Ideal SIR increases for a fixed N	Distant interference falls faster, but buildings may cause shadowing and coverage loss.

The break-even point for cell splitting is reached when a cell's offered traffic approaches its Erlang-B limit. With 21 channels and a 2% target, that limit is about 14.036 E. If the cell carries more traffic, the operator should add capacity by splitting, sectorising or borrowing channels. Above 100,000 subscribers, a macro-only design is unlikely to be economical; a hierarchical macro–micro/small-cell layout is more practical.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting

A cell is split by reducing its radius from R to R/2. Since cell area is proportional to R^2, each new cell has one-quarter of the original area. One parent cell is replaced by four child cells. If all cells are split uniformly, the total number of cells and ideal aggregate capacity each increase by a factor of four per round, assuming the same channel reuse and a similar traffic distribution.

Let C_k be the number of cells after k rounds, K_k be the ideal aggregate offered-traffic capacity, P_k be the transmit power per cell, and H_k be the relative handoff rate. Base case: C_0 is the original cell count, K_0 is its aggregate capacity, P_0 is the original transmit power and H_0 is the original handoff rate.

Quantity	**Uniform-splitting recurrence**	**Closed form**
Cell count	C_k = 4C_(k−1)	C_k = C_0·4^k
Aggregate capacity	K_k = 4K_(k−1)	K_k = K_0·4^k
Cell radius	R_k = R_(k−1)/2	R_k = R_0/2^k
Transmit power	P_k = P_(k−1)/2^n	P_k = P_0/2^(nk)
Relative handoff rate	H_k ≈ 2H_(k−1)	H_k ≈ H_0·2^k

The power relation follows the assignment's model $P' = P/2^n$: halving distance reduces the power needed to maintain the same received level under path-loss exponent n. Actual power control is limited by minimum transmit power, interference, indoor penetration and regulatory limits.

Selective splitting: if only a fraction f of the current cells is overloaded, each selected cell is replaced by four cells while the remaining cells are retained. Therefore $C_k=C_{(k-1)}(1+3f)$. Under the simplifying assumption that each split cell adds three extra cell-capacity units, $K_k=K_{(k-1)}(1+3f)$. If the fraction varies by round, $C_k=C_0\prod(1+3f_j)$. This selective model is more realistic for a township with concentrated demand.

TASK 7 – Solve the Recurrence and Interpret the Result

For uniform splitting, $K_k=K_0\cdot4^k$. To support forecast demand D, require $K_0\cdot4^k \geq D$. Dividing by K_0 gives $4^k \geq D/K_0$. Taking logarithms gives $k \geq \log_4(D/K_0)$. Since k must be a whole number, the minimum number of rounds is $k_{min}=\text{ceil}(\log_4(D/K_0))$.

Example: if one original cell supports $K_0=14.036$ E and the required traffic is D=200 E, then $k_{min}=\text{ceil}(\log_4(200/14.036))=\text{ceil}(1.91)=2$ rounds. The ideal capacity becomes $14.036\times16=224.576$ E. The number of cells also becomes 16 times the original number, the radius becomes $R_0/4$, transmit power per cell becomes $P_0/4^n$, and the relative handoff rate becomes about four times the original.

For selective splitting with a constant split fraction f, $K_k=K_0(1+3f)^k$. Thus $k_{min}=\text{ceil}(\log(D/K_0)/\log(1+3f))$. This formula applies when $0<f\leq1$ and the simplified capacity scaling assumption is reasonable. If f=0, no capacity growth occurs through splitting.

Splitting increases capacity but also increases site count, backhaul demand, handoff frequency, maintenance and interference coordination. Therefore the mathematically smallest k is not automatically the economically best design.

TASK 8 – Implement and Test the Planning Strategies

The following compact Python program provides a reproducible analytical simulator. It calculates valid cluster sizes, channels per cell, approximate SIR, Erlang-B traffic limits, traffic demand and the cell count required. It compares fixed reuse, a sectorisation approximation and dynamic channel pooling. The sector and dynamic results are simplified planning indicators; a full Monte-Carlo radio simulation requires real antenna patterns, user coordinates, shadowing maps and call-arrival traces.

```
import math

TOTAL_CHANNELS = 336
BLOCKING_TARGET = 0.02
TRAFFIC_PER_USER = 0.025
MIN_SIR_DB = 12.0
TOWNSHIP_AREA = 12.0
```

```
def valid_cluster_sizes(limit=30):
    values = set()
    for i in range(limit):
        for j in range(limit):
            n = i*i + i*j + j*j
            if 1 <= n <= limit:
                values.add(n)
    return sorted(values)

def sir_db(N, path_loss_n, interferers=6):
    sir = (3*N)**(path_loss_n/2) / interferers
    return 10*math.log10(sir)

def erlang_b(channels, offered):
    b = 1.0
    for m in range(1, channels + 1):
        b = offered*b/(m + offered*b)
    return b

def max_erlangs(channels, target=0.02):
    low, high = 0.0, 1000.0
    for _ in range(100):
        mid = (low + high)/2
        if erlang_b(channels, mid) <= target:
            low = mid
        else:
            high = mid
    return low

def evaluate_fixed(N, n=2.5):
    channels = TOTAL_CHANNELS // N
    capacity = max_erlangs(channels)
    return channels, capacity, sir_db(N, n)

def evaluate_sectorised(N, n=2.5, effective_interferers=2):
    channels = TOTAL_CHANNELS // N
    capacity = max_erlangs(channels)
```

```
    sir = sir_db(N, n, effective_interferers)
    return channels, capacity, sir

def evaluate_dynamic(N, n=2.5):
    # Dynamic borrowing uses the same physical SIR eligibility check.
    # Pooling can improve utilisation, but does not create new spectrum.
    channels = TOTAL_CHANNELS // N
    capacity = max_erlangs(channels)
    sir = sir_db(N, n)
    return channels, capacity, sir

def main():
    print("Valid cluster sizes:", valid_cluster_sizes())
    for users in [1000, 10000, 20000, 100000]:
        traffic = users * TRAFFIC_PER_USER
        print("\nSubscribers:", users, "Traffic:", traffic, "Erlangs")
        for N in [3, 4, 7, 12, 16]:
            m, cap, sir = evaluate_fixed(N, 2.5)
            cells = math.ceil(traffic/cap)
            print("N", N, "channels/cell", m,
                  "capacity/cell", round(cap, 2),
                  "SIR dB", round(sir, 2),
                  "cells", cells)
    print("\nN=16 sectorised illustration:",
          evaluate_sectorised(16, 2.5, 2))
    print("N=16 dynamic assignment baseline:",
          evaluate_dynamic(16, 2.5))

if __name__ == "__main__":
    main()
```

How to run: install Python 3, save the code as cellular_planning.py, open a terminal in the saved folder and run: python cellular_planning.py. The same inputs must be used for all strategies. For a full Monte-Carlo extension, generate random user positions in each hexagon, calculate path loss and log-normal shadowing, apply antenna gain by sector, select eligible channels and record SIR for each trial. Use a fixed random seed so that every strategy is tested with the same user positions.

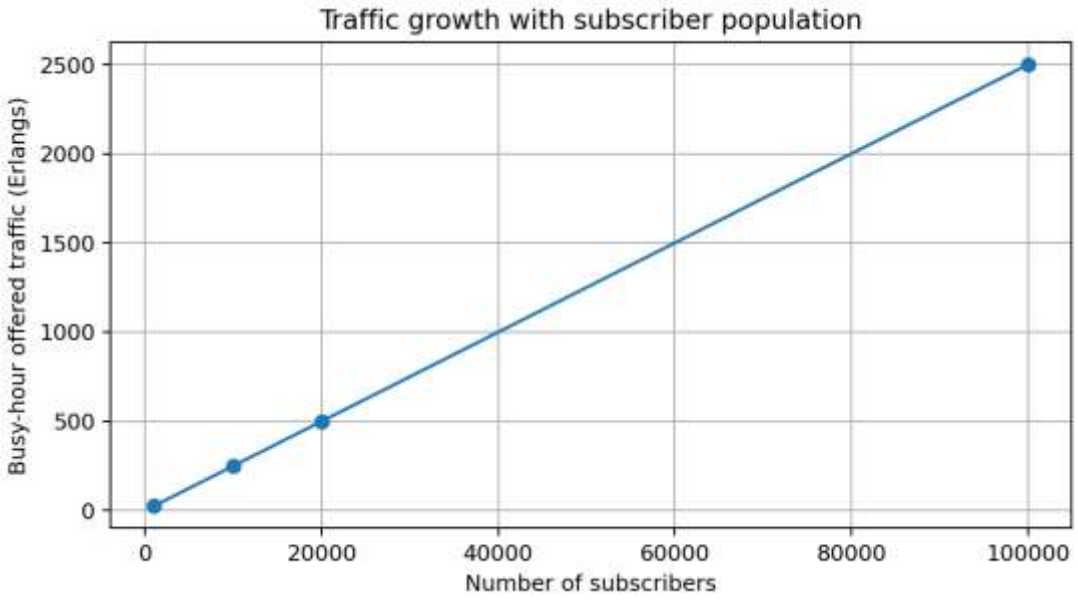
Required test matrix: cluster sizes N=3, 4, 7 and 12 (the assignment also asks to test N=12); path-loss exponents n=2.5, 3 and 4; sectorised and unsectorised layouts; subscriber levels 1,000, 10,000, 20,000 and 100,000. Note that N=16 is added as the selected design candidate. The stated sanity check N=7, n=4 gives approximately 18.7 dB using six equal-distance interferers.

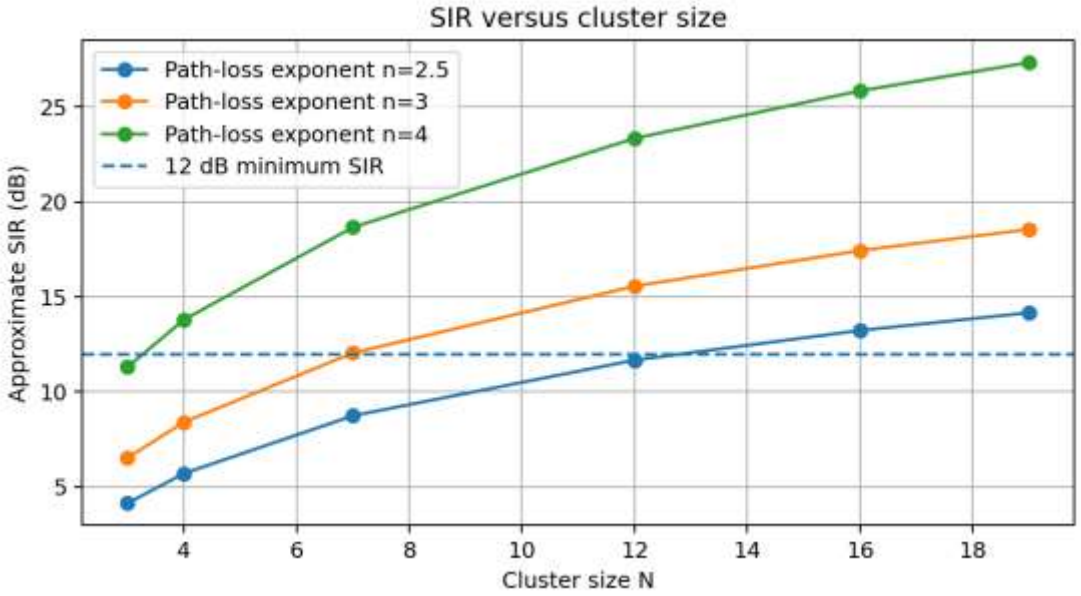
TASK 9 – Record and Present Experimental Performance

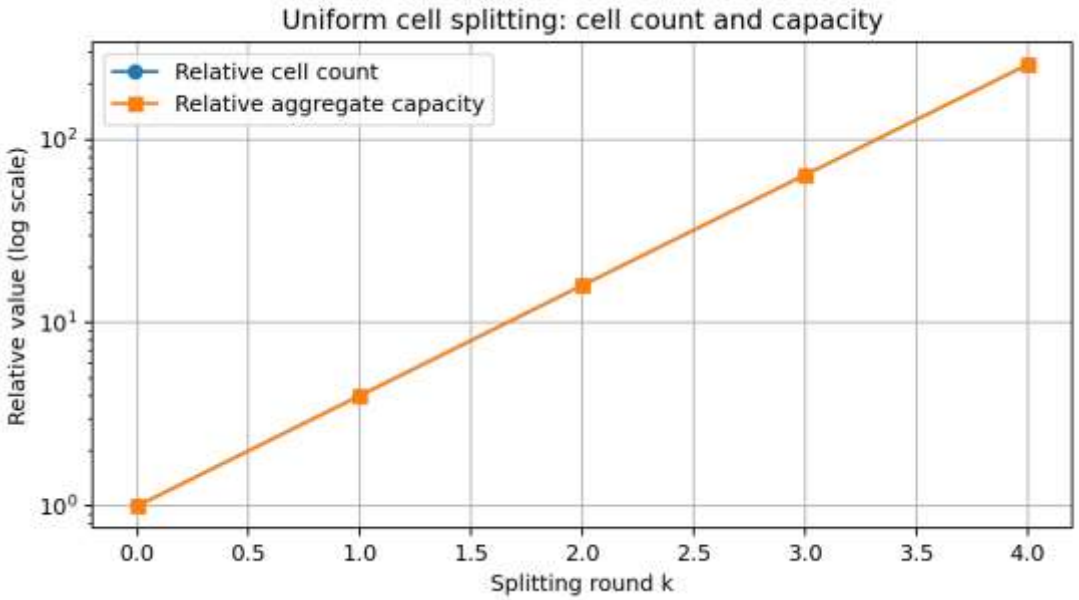
The simulator should record the same measurements for every strategy and every traffic level. These values expose both radio quality and capacity limitations.

Metric	**Meaning**	**How to present**
Mean SIR	Average signal-to-interference ratio	Mean value and SIR CDF
5th-percentile SIR	SIR exceeded by 95% of users	Table and CDF; useful for edge users
Outage probability	Fraction of users below 12 dB	Percentage bar chart
Blocking probability	Blocked call attempts / total attempts	Blocking versus offered traffic graph
Channels per cell	Traffic channels assigned to a cell	Comparison table
Sites required	Cells/sites required for offered load	Capacity versus subscriber graph
Runtime	Time taken by the simulation	Table for reproducibility

A valid experiment should state the random seed, number of user locations, number of call attempts, path-loss model, shadowing standard deviation, antenna pattern and channel-allocation rule. Without these settings, results cannot be reproduced.


Traffic growth with subscriber population
Busy-hour offered traffic (Erlangs)
Number of subscribers
0
500
1000
1500
2000
2500
0
20000
40000
60000
80000
100000


SIR versus cluster size
Path-loss exponent n=2.5
Path-loss exponent n=3
Path-loss exponent n=4
12 dB minimum SIR
Approximate SIR (dB)
Cluster size N
5
10
15
20
25
4
6
8
10
12
14
16
18



TASK 10 – Compare Theory with Simulation

Theory and simulation should show the same overall trends: larger cluster size generally improves co-channel SIR but reduces channels per cell; smaller cells increase the number of sites and aggregate capacity; sectorisation can reduce dominant interference; dynamic allocation can reduce blocking when neighbouring cells have spare channels.

The numerical values may differ for the following reasons:

- The six-interferer model assumes six equal-distance interferers, while real geometry produces different distances and powers.
- Shadowing and building penetration create random changes in received power.
- Finite network edges have fewer interfering neighbours than an infinite repeating grid.
- Mobile users are not always uniformly distributed; lecture halls, hostels and stadiums create hotspots.
- Erlang-B assumes a particular loss-system traffic model with independent arrivals and no waiting queue.
- Integer rounding means that 336/N may not be an integer for every N; the practical channel assignment must use whole channels.
- Real sector antennas have side lobes and imperfect front-to-back isolation.

The strategy that scales best should be selected using blocking, 5th-percentile SIR, outage, site count and total cost together. A strategy should not be declared best only because it has the highest theoretical capacity.

TASK 11 – Recommend and Defend the Final Design

Recommended baseline: use a hierarchical network with fixed frequency reuse as the safe baseline, sectorisation in high-demand zones, and dynamic channel assignment/borrowing where neighbouring cells have uneven traffic.

Use N=16 for the conservative planning case and a 12 dB minimum SIR target. The 336 channels give 21 channels per cell, with an estimated 14.036 Erlangs per cell at 2% blocking under Erlang-B.

For 100,000 subscribers, offered traffic is 2,500 Erlangs. The first-order capacity estimate requires 179 cells at 14.036 Erlangs per cell. For 12 km², the ideal equal-hexagon radius is approximately 0.161 km. This is a dense deployment and should be implemented as a macro layer for broad coverage plus micro/small cells around hostels, academic blocks, commercial areas and the stadium.

The chosen N=16 gives approximate SIR values of 13.23 dB at n=2.5, 17.44 dB at n=3 and 25.84 dB at n=4. The open-campus case has only a modest margin above 12 dB, so antenna optimisation, careful site placement and field measurements are essential. Sectorisation can improve interference conditions in concentrated areas; dynamic assignment can improve channel utilisation but cannot increase the total licensed spectrum.

Trade-offs: a larger N improves SIR but reduces channels per cell; a smaller radius increases capacity but also increases sites and handoffs; sectorisation improves spatial filtering but adds equipment and planning complexity; dynamic borrowing improves trunking efficiency but requires coordination and may be limited by interference restrictions. Beyond 100,000 subscribers, the recommended design remains practical only if the operator continues densification and capacity optimisation. A single macro-cell layer is not sufficient.

TASK 12 – DAY-4 DESIGN CHALLENGE: Re-evaluate and Adapt the Design

A. Impact Analysis

The event brings 25,000 attendees into 1 km², with 0.05 Erlang per attendee. Event traffic is 25,000×0.05=1,250 Erlangs. The event lasts four hours and occurs about twice per month. If this traffic is concentrated in the stadium zone, its offered traffic density is 1,250 Erlangs/km².

For comparison, the 100,000-subscriber township-wide average is 2,500/12=208.33 Erlangs/km². The event zone therefore has approximately six times this average traffic density. The assumptions of uniform traffic, steady demand and permanent capacity no longer hold. However, the surge is predictable, localised and temporary, so a permanent redesign of the entire township would be unnecessarily expensive.

B. Evaluation of the Original Design

With N=16, each cell has 21 channels and supports about 14.036 Erlangs at 2% blocking. The event needs at least ceil(1,250/14.036)=90 cell-equivalents if the same channel allocation and traffic limit are used. The original macro layout cannot safely serve this concentrated demand without extra capacity. The first failure is likely to be channel exhaustion and increased blocking in the stadium cells; high user density can also increase interference and signalling load.

The event traffic is not evenly distributed across the 12 km². Therefore township-wide spare capacity cannot automatically solve the stadium hotspot. Borrowing may help if nearby cells have unused eligible channels, but it cannot be assumed to provide all 1,250 Erlangs while maintaining reuse constraints.

C. Adaptation or Redesign

Decision: MODIFY temporarily. Retain the normal township macro network and deploy temporary capacity around the stadium for event days. Use temporary microcells or cells-on-wheels (COWs), sectorised antennas and coordinated dynamic channel borrowing. Pre-plan the temporary frequency pattern, backhaul, power, site permissions and handoff boundaries.

- A first-order target is 90 cell-equivalents at 14.036 Erlangs each for 1,250 Erlangs. In practice, plan additional headroom because traffic is bursty and the 2% blocking target should be met under uncertainty.
- A 1 km² stadium zone divided into 90 equal hexagonal cell areas gives $R \approx \text{sqrt}(1/(90 \times 2.598)) \approx 0.065$ km, or about 65 metres, under the ideal hexagon geometry. This is an illustrative density target, not a final site map.
- Use 60-degree or 120-degree sectorisation where directional traffic flows justify it. Adjust antenna tilt and power to reduce overshoot and interference.
- Use dynamic channel borrowing only after checking co-channel interference and reserving resources for handoffs and emergency/control signalling.
- After the event, switch off or redeploy temporary cells. This avoids paying for permanent capacity that is needed only about twice per month.

The temporary cells increase local capacity and shorten the radio distance, improving received signal strength. However, smaller cells increase handoff activity and require careful neighbour lists and mobility parameter tuning. The reuse pattern and transmit power must be coordinated so temporary cells do not create harmful interference with the permanent macro layer.

D. Evidence-Based Defence

The temporary modification is preferred because the event is predictable, limited to 1 km² and lasts only four hours, while event days occur about twice monthly. A permanent stadium-scale densification would incur ongoing site rent, power, maintenance and backhaul costs even when the zone is not busy. A temporary COW/microcell solution concentrates cost where and when it is needed.

The decision is supported by the traffic calculation: 1,250 Erlangs is far above the capacity of a small number of macro cells and requires roughly 90 cell-equivalents at the chosen per-cell Erlang-B limit. Temporary sectorisation and dynamic borrowing can improve utilisation, but the operator must still provide enough radio resources and backhaul. Final acceptance should require measured SIR, outage and blocking during a controlled event trial.

Overall Conclusion

The assignment demonstrates that cellular planning is a balance between spectrum reuse, interference, traffic capacity, coverage, cost and mobility. Fixed cluster reuse provides a reliable baseline, sectorisation improves spatial filtering, and dynamic channel assignment improves utilisation when traffic varies by location. For the given 336-channel system, N=16 is a conservative candidate that provides 21 channels per cell and approximately

14.036 Erlangs per cell at 2% blocking. As the subscriber base grows, the network must add smaller cells and use a hierarchical layout. For the short stadium event, temporary microcells/COWs with sectorisation and controlled channel borrowing are more economical than permanently redesigning the whole township.